

Kidney Cancer

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KEY POINTS

- Renal cell carcinoma (RCC) is diagnosed in approximately 64,000 patients/year in the United States.
- Clear cell renal cell carcinoma (ccRCC) is the most common histologic subtype.
- ccRCC is one of the relatively few malignancies that has an increasing incidence.
- RCC is frequently asymptomatic and is commonly diagnosed incidentally in the nephrology clinic during workup of acute kidney injury, obstruction, and/or hematuria.
- ccRCC is associated with paraneoplastic syndromes, most of which are related to its reprogrammed metabolism.
- Early immunomodulating therapies have shown minimal success, but newer targeted and checkpoint inhibitor treatments have improved survival.

Kidney cancer, among the 10 most common cancers in men and women, is one of the relatively few malignancies that has been increasing in incidence. This disease has proved refractory to most systemic therapies,¹ until quite recently, with the advent of immune checkpoint inhibitor antibodies,² and has a dismal prognosis, with an overall 5-year survival rate of 74%. This drops to 8% of those with metastatic disease (see <http://seer.cancer.gov/statfacts/html/kidrp.html>). Most importantly to the readers of this chapter, kidney cancer is the most common malignancy seen in the renal clinic.

Traditionally relegated to the practice sphere of urologists and oncologists, kidney cancer has recently been the subject of a resurgence in clinical and basic research (the latter in the area of metabolic reprogramming³), much of it undertaken by nephrology researchers. Thus it is imperative that practitioners of this subspecialty are keenly aware of the biology, risk factors, presentation, and management of this increasingly common disease, as has been recently reviewed.⁴ In this chapter, we discuss the basics of kidney cancer (also known as renal cell carcinoma, RCC), focusing on the most common type, clear cell RCC (ccRCC), so that the nephrologist will be comfortable with all aspects of a disease seen increasingly commonly in clinical practice.

BIOLOGY OF RENAL CELL CARCINOMA

RCC is the most common malignancy that originates from the renal cortex⁵ and is most often classified by histologic subtype. The most common subtype is ccRCC, comprising 75% to 85% of cases, but other histologic subtypes are also seen, many in children (Table 41.1). Each of the seven known genes that are mutated in the different types of kidney cancer are involved in a number of disparate pathways that regulate various aspects of cellular metabolism, such as oxygen and/or iron sensing, the tricarboxylic acid (TCA) cycle (also known as the Krebs cycle), glutamine metabolism, and tumor energetics.³ For this reason, kidney cancer has been aptly labeled a “metabolic disease.”^{6–8} Indeed, ccRCC has also been shown to be characterized by alterations in metabolism, as evidenced by nonstandard pathways in amino acid degradation, as well as energy production and protection from oxidative stress. This phenomenon of metabolic reprogramming⁹ was first described by Warburg early in the 20th century¹⁰ and has become evident in a variety of malignancies, including RCC. Such findings have been put to use in developing new biomarkers and therapeutic paradigms.³

ccRCC, the most common subtype and thus the primary subject of this chapter, comprises 70% of all RCCs and is one of the most lethal subtypes. The loss of the von Hippel-Lindau tumor suppressor (pVHL) gene¹¹ is common in ccRCC,^{12–14} which to a large degree dictates its biologic behavior. This gene loss leads to activation of hypoxia pathways, even in the absence of true hypoxia, and characterizes ccRCC as a malignancy true to the Warburg effect (i.e., aerobic glycolysis).¹⁰ Activation of downstream events by the VHL system, including neoangiogenesis and paraneoplastic phenomena, enables ccRCC cells to thrive as their surroundings become progressively more deprived of oxygen.¹⁵

RENAL CELL CARCINOMA: A METABOLIC DISEASE

ccRCC is an aggressive cancer that arises from the proximal tubular epithelium and, in its metastatic form, is associated with a high mortality. Each of the mutated genes for the various types of kidney cancer is involved in metabolic pathways, such as oxygen and iron sensing, the TCA cycle, glutamine metabolism, and tumor energetics. Studies involving different genomic platforms,^{16,17} also described in proteomic^{18,19} and metabolomic²⁰ studies, have identified a profound metabolic shift in aggressive ccRCCs involving the TCA, pentose phosphate, and phosphoinositide 3-kinase (PI3K) pathways, among others (Fig. 41.1). Furthermore, there are many reprogrammed pathways in ccRCC that have been, or can soon be, exploited for novel therapeutic approaches,

with the potential to transform the treatment of a disease with currently dismal options.

The hypoxia-inducible factor (HIF)–hypoxia pathway was one of the first pathways identified as being altered in ccRCC, but subsequent research has demonstrated that RCC has a number of additional metabolic abnormalities responsible for paraneoplastic syndromes. For example, the Warburg effect (aerobic glycolysis), arginine synthetic abnormalities (due to arginosuccinate synthetase-1 [ASS1] deficiency), and glutamine pathway reprogramming are all pronounced in RCC.^{18,21,22} The elucidation of these and other metabolic abnormalities turns out to be a gold mine for therapeutic exploration. Targeted therapy using pathway-specific inhibitors, which in some cases are compounds that were discarded after their discovery, would show high specificity, with fewer adverse effects.²³

BIOLOGY AND RATIONALE OF CURRENT THERAPEUTICS

Prior therapeutic approaches exploited the high level of immunogenicity of RCC and used immunotherapy with interferon and interleukin-2 (IL-2), but these were associated with severe and unpleasant adverse effects and only modest success. More recently, therapies targeting newly elucidated biochemical pathways have a better response, as well as fewer adverse effects, and there are even more pipeline therapies based on metabolic reprogramming, as with tryptophan²⁴ and arginine²¹ reprogramming. Most recently, the immune checkpoint inhibitors have shown considerable promise in treating ccRCC, and studies are currently underway to find optimal combinations using these new drugs.² However, the marked inter- and intratumoral heterogeneity in ccRCC²⁵ has made it difficult to study this disease as a single entity with respect to therapeutic response.

RENAL CELL CARCINOMA HISTOLOGY AND GENETICS

The traditional classification of RCC is based on histomorphologic criteria that divide RCC into the three

Table 41.1 Subtypes of Renal Cell Carcinoma

Subtype	Incidence
Clear cell	70%–88% of RCCs ^{212–214}
Papillary type 1	Papillary types 1 and 2 together constitute
Papillary type 2	10%–20% of RCCs ^{214–216}
Chromophobe	5% ^{7,215}

ccRCC, Clear cell renal cell carcinoma; RCC, renal cell carcinoma.

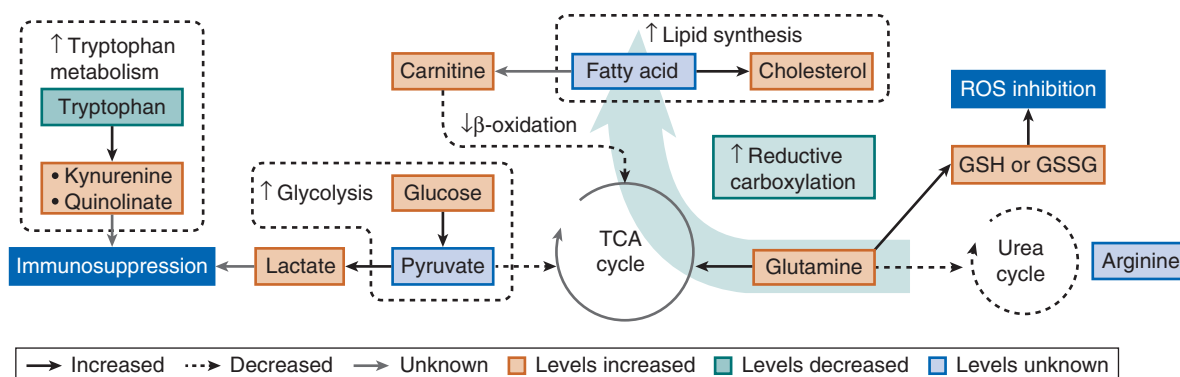


Fig. 41.1 Metabolic reprogramming in clear cell renal cell carcinoma (ccRCC). In ccRCC, aerobic glycolysis, carnitine and lipid synthesis, reductive carboxylation, the glutathione oxidized glutathione (GSH/GSSG) pathway, and tryptophan metabolism are upregulated, whereas the urea cycle and energy production through the tricarboxylic acid (TCA) cycle are downregulated. These changes are advantageous for ccRCC cells because they enable them to survive in conditions of nutrient depletion and hypoxia, provide the cellular building blocks required for proliferation, and result in the production of immunosuppressive (kynurenine and quinolinate) and antioxidant metabolites (GSH and GSSG). GSH, Glutathione; GSSG, glutathione disulfide; ROS, reactive oxygen species.

main types—clear cell RCC (ccRCC), papillary RCC (pRCC), and chromophobe RCC (chRCC; Table 41.1). Recent advances in the genomic profiling of RCC have resulted in further characterization of tumor types, allowing for better understanding of their biology and greater specificity of treatment targets.

CLEAR CELL RENAL CELL CARCINOMA

ccRCC is the most common type of renal cancer and originates in proximal tubular cells. Genetic abnormalities in ccRCC are roughly divided into those affecting the VHL gene, epigenetic regulators and chromatin remodeling genes, and disruption of PI3K-AKT-mTOR signaling.^{26,27}

The large majority of cases of ccRCC are sporadic, and only 2% to 3% of ccRCC cases are linked to hereditary diseases. Germline mutations in the pVHL gene on chromosome 3p25²⁸ lead to the formation of benign or malignant tumors and to the formation of cysts in other organs. Conversely, mutations or silencing of the same VHL gene are associated with more than 80% of sporadic ccRCC cases.^{29,30} VHL disease is relatively common, with an estimated incidence of 1 in 36,000 live births,^{31,32} and has a penetrance of more than 90% by age 65 years.³³ Most subjects affected by the disease inherit a germline mutation of the gene from the affected parent and a normal gene from the unaffected parent. All the cells of affected individuals who inherit the genetic trait will have germline mutations of VHL. However, tumors will develop only in those cells that undergo a deletion or mutation of the remaining wild-type allele (the so-called second hit) and, in addition, are constituents of vulnerable target organs, such as kidneys, central nervous system (CNS), adrenal glands, pancreas, and reproductive adnexal organs.

The VHL gene has three exons and is located on the short arm of chromosome 3 (3p25-26).³⁴ Since its discovery, it has been well established that this gene plays a crucial role in a variety of cellular pathways. The most important function of VHL protein (pVHL) is polyubiquitination of a variety of proteins, including HIF 1 and 2. This process marks them for degradation in the presence of normal tissue oxygenation.³⁵ However, under hypoxic conditions, HIF activates the transcription of hypoxia-inducible genes that results in the enhanced production of vascular endothelial growth factor (VEGF), erythropoietin, platelet-derived growth factor- β (PDGF- β), transforming growth factor- α (TGF- α), and several glycolytic enzymes.^{36–38} Failure of pVHL to rein in the activity of HIF in normoxia has crucial implications for the clinical behavior and metabolic features of ccRCC. Among them are erythropoietin-dependent erythrocytosis, marked angiogenesis, and cell proliferation mediated by TGF- α and VEGF,^{36,39} as well as aerobic glycolysis (Warburg effect) and the immunosuppressive effects of increased tryptophan catabolism.⁴⁰ The nature of interactions between HIF and cellular metabolism is likely bidirectional, as is evidenced by the inhibitory effects of the gluconeogenic enzyme fructose-1,6-bisphosphatase on HIF function.⁴¹ Although some studies have shown an association between such metabolic abnormalities and tumor aggressiveness, nontargeted metabolomics analyses have not yet become a part of clinical practice.⁴²

The second most common type of mutation involves several epigenetic regulators also localized on 3p25 chromosome, including polybromo 1 (PBRM1), BRCA-associated protein-1

(BAP1), and SET domain-containing 1 (SETD1) genes involved in regulation of chromatin maintenance and remodeling. Generally, hypermethylation of promoter sites is associated with the tumors carrying these mutations (with the exception of SETD1) and higher tumor grade, but their exact role in ccRCC pathogenesis is not well understood.^{43–47}

Genetic aberration of mTOR pathway proteins (PTEN, AKT, PIK3CA, and mTOR) involved in cell proliferation signaling have been identified in up to 28% of RCC patients.⁴⁸ Deficiency in PTEN, in particular, is associated with more aggressive tumors.⁴⁹ Together with the abnormal expression of the focal adhesion kinase modulator of the mTOR pathway, these genetic defects are present in more than 50% of ccRCC patients.⁴⁸ As part of the Cancer Genome Atlas project, less frequent mutations of genes involved in cell metabolism and division have been described, with some of them, such as p53 overexpression, associated with a worse prognosis.⁴⁸

PAPILLARY RENAL CELL CARCINOMA

This second most common RCC is also of proximal tubule origin, but it has been studied less extensively than the other subtypes.⁵⁰ The type 1 papillary RCC (pRCC) is characterized by MET proto-oncogene-activating mutations taking place in germ cells in the hereditary form of the disease and 10% to 20% of cases of somatic mutations in the most common and sporadic form of the disease.⁵¹ Additionally, trisomy 7 has been implicated in MET amplification in some cases of pRCC.⁵² Type 2 pRCC is linked to activation of the NRF2 antioxidant response element pathway as the result of augmented oxidative stress. Mutations in the fumarate hydratase (FH) gene in this disorder are associated with hereditary leiomyomatosis and renal cell carcinoma (HLRCC).^{29,53} Mutations in the epigenetic modifier genes PBRM1, BAP1, and SETD1 have been described with this tumor type, but seem to be less prevalent compared with ccRCC.⁵² Type 2 pRCC portends a considerably worse prognosis compared with type 1 pRCC.^{54,55}

CHROMOPHOBE RENAL CELL CARCINOMA

This rare cancer that originates from the collecting duct and that is similar to benign oncocytomas comprises less than 5% of RCC and is often linked with whole-chromosome losses and germline mutations in the folliculin gene (FLCN) found in the autosomal dominant Birt-Hogg-Dubé syndrome.^{56,57} The most frequent mutation in sporadic cases of chRCC involves downregulation of p53 tumor suppressor signaling and loss of function of PTEN leading to PI3K-driven cell proliferation. ChRCC is more common in younger females and is typically the least aggressive of all RCC types, unless characterized by sarcomatous transformation.⁵⁸

PRESENTATION

The unusual and pleiomorphic presentation of RCC is reflective of its biologic characteristics, with clinical presentation unlike that of most other urologic tumors, and this challenge is reflected in its well-deserved appellation “the internist’s tumor.”⁸ Thus, the diagnosis can be challenging due to its nonspecific and often systemic symptoms.⁸

The clinical presentation of RCC is variable; many patients have clinically occult disease and often present with a late and more advanced stage at the time of diagnosis. About 25% of patients have metastatic disease or advanced local

disease at the time of presentation.^{59–61} The classic presentation of hematuria, flank pain, and fever is seen in less than 10% of patients,⁶² and more than 50% of patients are now diagnosed based on incidental findings in abdominal radiologic imaging performed for unrelated reasons.⁶³

RCC patients can also present with nonspecific clinical findings, including hypertension (HTN), weight loss, malaise, night sweats, and the new onset of a varicocele. Paraneoplastic syndromes, including fever, anemia, hypercalcemia, erythrocytosis, and abnormal liver enzyme levels not due to metastatic spread (Stauffer syndrome), are seen in 7% of cases; polyneuropathy and amyloid A amyloidosis occur in 3% to 5%.⁶⁴ Fever occurs in up to 20% of patients and is often accompanied by night sweats, anorexia, weight loss, and fatigue.⁶⁴ Hypercalcemia occurs in up to 15% of cases and has been identified as a negative prognostic factor, sometimes associated with lytic bone metastases.⁶⁰ The ectopic production of parathyroid-related peptide by the tumor has been linked to hypercalcemia in the absence of bone metastases.^{65–67} Erythrocytosis has been described in 1% to 5% of patients with RCC and is thought to be secondary to the unregulated production of erythropoietin by the cancer cells.^{68,69}

RCC typically presents at 60 to 70 years of age. Other RCC risk factors in the general population include male gender, obesity, tobacco use, diabetes mellitus (DM), and HTN.⁷⁰ The incidence of RCC has increased more rapidly among blacks than whites, with a shift from a predominance among whites to blacks.⁷¹ Although RCC has a higher incidence among those with obesity, this population tends to have better survival, which may be attributed to the potentially protective effect of brown fat. Furthermore, alcohol exposure paradoxically seems to lower RCC risk, unlike with other tumors, such as liver or colon cancer.^{72,73} Cystic disease, which may progress to (or hide) RCC, is associated with end-stage renal disease (ESRD), and prolonged dialysis dependency also predispose ESRD patients to renal cancers, with a 100-fold increased risk.⁷⁴ A similar association has been made with chronic kidney disease (CKD), but to a lesser degree than with ESRD.⁷⁵

CYSTIC DISEASES AND RENAL CELL CARCINOMA

The link between cystic diseases and RCC was described in 1880 by Brigid and Severi.^{76,77} Cortical cystic disease can range from simple cysts to complex cysts, with increasing risk for malignancy, as defined by the Bosniak classification.^{78–80} However, CKD-related specific cystic diseases, including acquired cystic kidney disease (ACKD) and polycystic kidney disease (PKD), have different characteristics from those of the general population and deserve further consideration, as discussed later.

ACQUIRED CYSTIC KIDNEY DISEASE AND CHRONIC KIDNEY DISEASE

ESRD is associated with a high risk of RCC.⁸¹ The incidence of RCC in the ESRD population ranges from 1% to 7% in various studies, which is reportedly exponentially greater than that in the general population.^{82–86} Malignant transformation

from ACKD that is seen primarily in the ESRD population appears to account partially for this association.⁸¹

The development of ACKD has been described among 7% to 22% of CKD patients, but this escalates among dialysis-dependent ESRD patients (10%–44% within 1–3 years) and increases further with prolonged duration of dialysis (>90% after 5–10 years).^{84,87–89} Although the presence of ACKD leading to RCC has also been described among transplant recipients (23%), this is far less than that observed in ESRD patients on dialysis (80%).⁸⁴

In addition to ESRD duration, risk factors for development of ACKD and likely progression to RCC include male gender, younger age at ESRD onset, and diagnosis of glomerulonephritis; however, diabetic nephropathy has been associated with a lower incidence of ACKD among Japanese dialysis patients. Furthermore, kidney volumes and the rate of volume increase appear greater in men compared with women. Other demographic factors, including race, have no association with ACKD.^{84,87–89} Dialysis modality is not consistently related to ACKD risk^{84,87}; however, in some reports, ACKD was less common with the peritoneal dialysis modality.^{82,87,90} Much of the literature on ACKD and RCC was reported in the Japanese population, in whom transplantation is far less common, possibly because of cultural reasons. Therefore, the dialysis vintage of these ESRD cohorts is much greater than that in the United States, and these associations vary according to specific populations.

The diagnosis of ACKD is made by the presence of more than three cysts collectively in both kidneys.^{87,88} ACKD cysts tend to be smaller in size (typically <0.6 cm in diameter, but can range up to 2–3 cm) compared with cysts in PKD or other cystic diseases. Symptoms related to ACKD are rare, except when related to complications. Cyst rupture with hemorrhage, cyst infections, erythrocytosis, calcium oxalate deposition, and stone formation has been described.^{87,91}

It has been theorized that the acquired cysts, unlike those seen in PKD, communicate with the renal tubular system and therefore retain some normal functional properties. Cyst fluid content has been found to be higher in creatinine content, suggesting some filtering capacity, unlike simple or PKD cysts (which have similar cyst and plasma creatinine levels). This may explain why cysts often regress with ACKD after renal transplantation when normal filtration is restored.^{87,92}

The proliferation of cysts may stem from a reparative response to uremic metabolites, chronic acidosis, and ischemia.^{82,87} Growth factors from cyst fluid have been proposed to induce cellular hypertrophy and hyperplasia in cell culture^{82,93}; this proliferative effect was also observed in vitro with a polypeptide (15–30 kDa) isolated in plasma from dialysis patients.⁹⁴ Other potential causative factors include dialysis-related factors; however, these do not account for ACKD observed in nondialysis CKD patients. Data supporting these findings were published decades ago, without revealing a clear pathogenesis to date.^{82,87}

ACQUIRED CYSTIC KIDNEY DISEASE AND RENAL CELL CARCINOMA

The close association of ACKD with both RCC and ESRD may explain the increased risk of RCC in ESRD.⁸⁴ The annual incidence of RCC in a Japanese ACKD cohort with 20-year follow-up was 0.151%/year for those on dialysis less than 10

years and 0.340% with dialysis duration longer than 10 years,^{87,89} which is higher than what has historically been observed in the general population (0.008%)⁹¹. Furthermore, the finding of shared risk factors between ACKD and RCC (male gender and ESRD) and related pathologic findings support the likelihood that ACKD can evolve to RCC.⁸⁷ Acquired cysts seen in dialysis patients have similar immunohistochemical profiles and histologic traits (kDaepithelial cell-lined cysts, with eosinophilic or foamy cytoplasm) compared with the RCCs associated with ACKD, suggesting that acquired cysts are precursor lesions to RCC in this population.^{88,95}

Papillary RCC is the predominant pathologic subtype among ACKD and dialysis-associated tumors, in contrast to the sporadic form of RCC, ccRCC, typically seen in the general population. Distinctive immunohistochemical patterns of expression in RCC specimens appear to differentiate papillary RCC in ACKD-related cancer from ccRCC among Japanese ESRD patients. In this population, although hypoxia-inducible gene-2 (HIG2) protein expression was increased in both types of RCC, HIG2 was particularly higher with papillary RCC, whereas HIF-1 α and activated nuclear factor-kappa B (NF- κ B) elevation was seen primarily with ccRCC. Nontumorous but cystic regions had increased expression of all three proteins. HIG2 and HIF1- α were particularly upregulated with hyperplastic cysts. Persistent exposure to these mitogenic factors in both tubulointerstitial injury and RCC has been proposed to explain the carcinogenesis of ACKD to RCC.⁹⁶

POLYCYSTIC KIDNEY DISEASE AND RENAL CELL CARCINOMA

The prevalence of RCC in autosomal dominant polycystic kidney disease (ADPKD) does not appear to be greater than that in the general population, according to small case series and observational studies,^{97–101} although there is some controversy in the literature on this subject. As in the general population, the histologic diagnosis of RCC in ADPKD tends to be clear cell carcinoma in the most current case series to date,^{97,99,101} although in one European series, tubulopapillary pathology was prominently observed as well (42%).¹⁰⁰ RCC in ADPKD presents at a younger mean age (50–60 years) than spontaneous RCC, but often with advanced disease, where one-third of patients have bilateral kidney involvement or metastatic disease. Presentation with advanced disease may be caused by delayed diagnosis, given the complexity of diagnostic images in the presence of multiple benign cysts in ADPKD. Symptomatic diagnosis rather than incidental discovery of RCC is more common in this select population.^{97,99,101}

PROGNOSIS OF RENAL CELL CARCINOMA IN THE END-STAGE RENAL DISEASE POPULATION

The prognosis of RCC in the ESRD population is equivalent or better compared with that in the general population.^{102–106} The 5-year survival from RCC related to ACKD was comparable between the two groups,¹⁰⁶ but a broader examination of RCC related to all causes in a contemporary French cohort has revealed that 5-year survival is 90.1% for the ESRD group ($n = 303$) compared with 69.0% in the non-ESRD group ($n = 947$). Cancer-specific mortality was much lower, at 4.3%

(vs. 27.6%) for the ESRD group also. About one-third (30%) of the ESRD patients had ACKD, and RCC was incidentally diagnosed in most ESRD patients (87%). The ESRD group had more favorable characteristics, including younger age, better performance status, fewer symptoms, smaller tumor size (3.7 vs. 7.3 cm), and lower tumor grade and stage (10% vs. 42% with stage ≥ 3 tumor), and patients were more likely to have papillary tumor (37% vs. 7%) than observed in the general population.¹⁰⁵ Similar findings have been replicated in other studies.^{102,104,105} Higher survival in the ESRD group is attributed to the incidental finding, presumably leading to earlier diagnosis compared with the non-ESRD group.¹⁰⁵ Further examination of a Japanese dialysis cohort has revealed that those with symptomatic (vs. incidental) diagnosis of RCC are more likely to have advanced tumor features, including greater size, higher stage, higher grade, and lower cancer-specific (76.9% vs. 95.3%) and overall (64.2% vs. 84.9%) 5-year survival. In addition to these features, longer dialysis duration and diabetic nephropathy were also identified as prognostic factors in a multivariable analysis for survival.¹⁰³ The less aggressive course of non-ccRCC may also explain the better outcomes associated with ESRD-related RCC.¹⁰⁴

RENAL CELL CARCINOMA RISK AND KIDNEY FUNCTION

The causal relationship between RCC and CKD is complex, with each pathology conferring risk to the other.⁴ The 100-fold increased risk of RCC among ESRD patients⁸⁶ may be due to the higher prevalence of ACKD, but this does not clearly explain the association of pre-ESRD CKD with higher risk of RCC.⁷⁵ The largest cohort study to date has examined the risk of RCC due to CKD stratified by estimated glomerular filtration rate (eGFR) among 1,190,538 subjects without known RCC. With decreasing eGFR ranges, RCC risk increased as follows: stage 3a CKD, hazard ratio (HR), 1.39 (confidence interval [CI], 1.22–1.58); stage 3b, HR, 1.81 (CI, 1.51–2.17); and stages 4 and 5 CKD, HR, 2.28 (CI, 1.78–2.92).⁷⁵ The association was specifically with clear cell carcinoma and not with the other RCC subtypes. In contrast, the predominant RCC subtype seen with ESRD is pRCC, suggesting pathology that is unrelated to ACKD. Moreover, even early findings of kidney pathology, such as albuminuria, have been related to cancer, including RCC.¹⁰⁷ The link between kidney function and localized RCC risk was abundantly apparent among 202,195 renal transplant recipients. During periods of graft failure resulting in ESRD, the incidence of RCC rose and, during periods of graft function, RCC incidence fell (Fig. 41.2). This pattern recurred during periods of kidney function loss (repeat graft failure).¹⁰⁸ How loss of kidney function is related to RCC risk is largely unknown, but it has been attributed to inflammation, acidosis, or uremic factors often implicated in advanced CKD.^{4,75,107}

ACQUIRED CYSTIC KIDNEY DISEASE, RENAL CELL CARCINOMA, AND RENAL TRANSPLANTATION

Screening for RCC in renal transplant candidates who have ESRD is controversial and is not currently recommended,

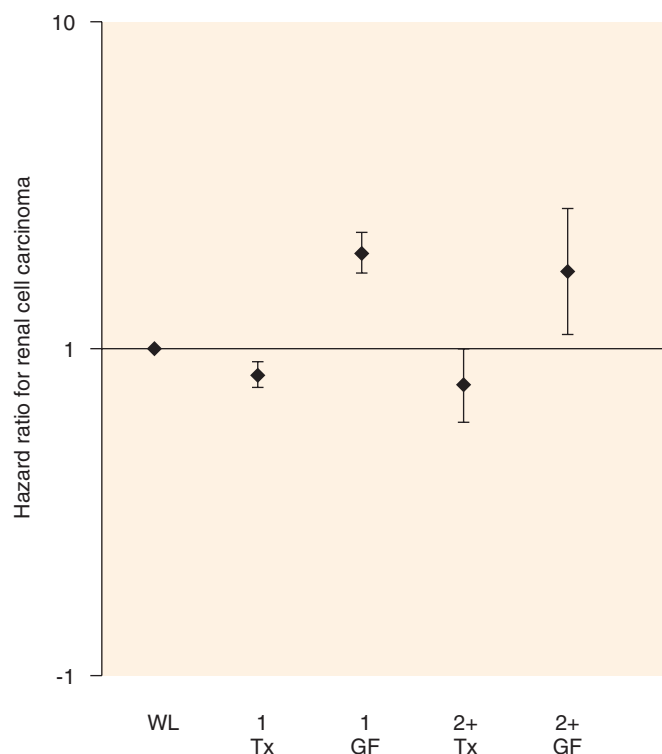


Fig. 41.2 Risk of renal cell carcinoma (RCC) according to periods of renal function. Shown are hazard ratios (HRs) comparing successive intervals for cancers with distinct alternating incidence patterns. Each point represents the HR comparing each interval with the immediately preceding interval. No HR estimate is given for the last interval when no cancers were observed in that interval. Vertical error bars represent end-stage renal disease–related cancers (95% confidence interval). Results are shown for cancer types that have demonstrated alternating patterns of cancer incidence across intervals—that is, based on the point estimates for the HRs, regardless of their statistical significance. GF, Graft failure; Tx, transplant; WL, waiting list. (Modified from Yanik EL, Clarke CA, Snyder JJ, et al. Variation in cancer incidence among patients with ESRD during kidney function and nonfunction intervals. *J Am Soc Nephrol*. 2016;27(5):1495–1504.)

despite the use of powerful immunosuppression drugs that increase cancer risk in general.^{84,109,110} The question remains whether the risk of RCC is increased among transplant recipients, given the higher incidence of ACKD with ESRD. Most cohort studies have found that the prevalence of native kidney RCC for renal transplant recipients, ranging from 3% to 5%, was no different from the nontransplant ESRD population; however, this is still more than 100-fold greater than that of the general population.^{86,111–113} On the other hand, donor allograft to the recipient transmission of RCC is rare (2/10,000 organs transplanted), with excellent outcomes, as reported in a meta-analysis but will not be explored further here.^{114,115}

ACKD is less commonly described among transplant recipients (23%–33%) in most cohorts^{86,111,116} but has been reported as high as 57%,¹¹⁷ which is far less than that observed in ESRD patients on dialysis (80%).⁸⁴ The close association of ACKD and RCC is still apparent in the renal transplantation population; ACKD is higher among RCC transplant patients than among the overall, non-RCC transplant patients (60%–90% vs. 23%–33%, respectively).^{86,113,117} However, there

was an exception in one study, where they were found to be equivalent (33% with RCC vs. 29% without RCC).¹¹¹ Reciprocally, more transplant patients with ACKD are likely to have RCC than those without ACKD (11%–19% vs. 0.5%–0.6%). Generally, transplant recipients with RCC have favorable characteristics—younger age, smaller tumor size, less metastatic disease, more with stage T1a, and papillary subtype diagnosis—than nontransplant ESRD patients.¹¹⁷ One transplant-related factor, a higher humoral rejection rate, has been observed in those diagnosed with RCC.¹¹¹

Survival outcomes for transplant recipients with RCC have varied according to particular analyses, but generally appear to be more favorable compared with those of the general population, and possibly the dialysis population, likely due to closer surveillance of these patients. The 10-year, cancer-specific survival of 88% to 95% in transplant recipients is higher than that in the general population (75%).¹¹⁶ Additionally, 97% of transplant recipients achieved a cancer-specific survival of 5 years, better than 77% in the nontransplanted ESRD group in one study.¹¹⁷ The overall patient and graft survival between those with and without RCC have been comparable.¹¹¹

Transplant recipients are a self-selected healthier cohort who, despite RCC prevalence similar to their dialysis counterparts and persistent association with ACKD, appear to have higher oncologic and overall survival than other ESRD patients and the general population. Furthermore, decision analysis did not support screening for ACKD in the entire ESRD population, given the minimal gain of life expectancy; however, the authors concluded that in younger healthier patients, screening may be considered.^{109,110} Clinical judgment and risk analysis should be used to determine the benefit of screening on an individual basis.

EVALUATION AND PROGNOSTIC ASSESSMENT OF RENAL MASSES

Renal masses are increasingly being discovered spontaneously with the frequent use of diagnostic imaging in clinical practice, in which the rate of incidental diagnosis increased from about 10% in the 1970s to 60% less than 3 decades later.¹¹⁸ Incidental diagnosis has led to earlier detection of RCC at lower stages. To distinguish suspicious masses from benign tumors, clinicians rely on cyst classification systems, such as the Bosniak classification, which is based on cyst wall characteristics (thin vs. thickened), number of septae, calcification, contrast enhancement, and size (≥ 3 cm). Classes I and II define relatively benign lesions, starting from a simple cyst; then classes IIF and III have progressively increasing complexity, leading to class IV, composed of frankly malignant lesions (Table 41.2).^{78–80} The malignancy risk of most cystic lesions can be assessed using these criteria. Prognosis is further determined with staging and grading tools, which are often incorporated into integrated staging systems for enhanced differentiation.^{119,120}

RADIOLOGIC EVALUATION OF RENAL MASSES

Along with clinical assessment, diagnostic imaging has become the mainstay of renal mass evaluation. The primary choice of imaging modality for screening and diagnosis of renal

Table 41.2 Bosniak Classification of Renal Cystic Masses

Classification	Description	Likelihood of Malignancy (%)
Bosniak I	Simple cyst of water density with a thin wall that does not contain septa, calcifications, or other solid components	1.7
Bosniak II	Cyst may contain a few hairline septa and fine calcifications; uniform high-attenuation lesions <3 cm that do not show contrast enhancement	18.5
Bosniak IIF	Cysts may contain multiple hairline thin septa or minimal smooth wall thickening; calcifications may be thick and nodular, but no contrast enhancement	>18.5
Bosniak III	More complex “indeterminate” cystic lesions, with more calcification and more prominent septation; thicker irregular or smooth walls; measurable contrast enhancement present	33
Bosniak IV	Clearly malignant cystic masses with all the criteria of category III but also contain contrast-enhancing soft tissue components adjacent to, but independent of, cyst wall or septum	92.5

masses is computed tomography (CT) due to its high level of detection, particularly as higher-resolution multidetector CT scanners have come to be used more in most medical imaging centers.^{118,121} Renal tumor detection by CT has high sensitivity (100%) and specificity (95%), and CT scanning can determine tumor size within 0.5 cm of the actual tumor measured histopathologically.¹¹⁸ In a study examining the accuracy of tumors sizes measured by CT, 85% of renal masses identified by CT of 133 specimens were pathologically confirmed, although the tumor sizes were slightly overestimated by CT in this retrospective study.¹²² Accuracy of staging by multidetector CT also excels in regard to extrarenal disease assessment, with high specificity, particularly for detection of capsular invasion (85%), perirenal fat invasion (98%), lymph node (82%), renal vein (98%), inferior vena cava (98%), and adrenal gland (98%) involvement,^{118,121} and high sensitivity of 84% to 100%, except for capsular invasion (68%), perirenal fat invasion (46%), and lymph node involvement (77%).^{118,121} Furthermore, RCC subtypes are identifiable on CT based on vascularity or heterogeneity of the mass, because ccRCC is more likely to be hypervascular and has heterogeneous parenchyma.¹²¹

Ultrasound is also commonly performed during the course of routine medical care, which has led to incidental finding of renal masses. It is particularly useful for distinguishing noncystic tumors from simple and complex cysts and equally useful for predicting T stage.¹²¹ However, ultrasound is associated with inferior sensitivity and specificity for the detection of small tumors (1–2 cm) and is therefore a poor screening tool. Although detection rates for both ultrasound and CT for tumors are poor when evaluating masses smaller than 0.5 cm (0% and 47%, respectively), CT remains far superior to ultrasound for 1- to 2-cm lesions. Detection improves with increasing size, from 20% for 1-cm lesions to 79% for 2-cm lesions, but remains suboptimal until tumor size approaches 3 cm, where the detection rate by ultrasound (100%) finally reaches equivalence to CT.¹²³ Once the lesion is identified, ultrasound can be used for repeated monitoring, especially when avoidance of iodinated intravenous contrast is preferable.¹²¹

Novel contrast-enhanced ultrasound using microbubbles of injectable gas has been found to be even more sensitive than CT.¹²¹ Early generations of these microbubbles were nonnephrotoxic but riddled with problems due to instability in the circulation, leading to cardiopulmonary complications and, in some cases, death. The current microbubbles, now

smaller and more stable, have been approved for use by the U.S. Food and Drug Association (FDA).¹²⁴

Magnetic resonance imaging (MRI) is traditionally considered to be the best modality for evaluating soft tissue pathology, including RCC, with optimal visualization of hemorrhage, micro- and macroscopic fat, and intracystic structures. The sensitivity and specificity for differentiating localized disease from the next stage (T3) is higher than any other modality (84% and 95%, respectively; positive predictive value of 96%) and 100% for both in detecting inferior vena cava (IVC) involvement.¹²⁵ In an advanced CKD or ESRD population, however, the use of MRI is excluded due to gadolinium-induced nephrogenic systemic fibrosis.¹²⁶ Although MRI remains highly effective in renal mass characterization and staging, the newer multidetector CT scanners have achieved an equivalent level of accuracy and detection rate as compared with MRI. Therefore, ultimately, CT has become the primary imaging modality to aid in the management of RCC.¹²⁷

Because many ccRCCs are glutamine-avid and glutamine reprogramming has been identified in ccRCC,^{3,20} this amino acid has been exploited in an experimental setting for novel positron emission tomography (PET)-based imaging techniques. A series of reports have suggested that ¹⁸F-fluorodeoxyglucose (¹⁸F-FDG)-negative tumors might use glutaminolysis preferentially to glycolysis.^{128–131} Concomitantly, we and others have found that RCC is strongly glutamine-avid and possibly even glutamine-addicted.^{20,132} As a result of this work, the glutamine analog 4-¹⁸F-(2S,4R)-fluoroglutamine was developed and shown to be taken up by cancer cells.¹³³ PET imaging has now been used to show glutamine uptake in animal models and in patients with glioma.¹³⁴ Moreover, glutamine uptake correlated with disease progression in these patients. In ccRCC, we have used PET scanning to show glutamine uptake in vitro and in several mouse xenograft models^{134a} so that PET-based imaging based on the glutamine reprogramming of RCC could potentially be used in the future for RCC staging, patient selection, and real-time monitoring of glutaminase inhibition as a novel therapy for this disease.³

DIFFERENTIATION OF BENIGN TUMORS FROM RENAL CELL CARCINOMA

The most commonly observed benign lesions are angiomyolipomas, which are composed of blood, smooth muscle, and

adipose tissues. They are primarily seen among middle-aged women, 40 to 60 years of age.¹³⁵ Angiomyolipomas tend to be adipose-rich tumors that are easily distinguished from RCC, except when they are nonfatty (4.5%), which would require further assessment with MRI.^{135,136} Oncocytomas are the next most frequently found benign renal mass (3%–5% of case series) generally seen among older men (aged in their 70s). These tumors are from tubular cells in the cortical collecting duct and can be difficult to differentiate from RCC due to their similar rate of growth.^{135,136}

When current imaging techniques cannot adequately determine whether a tumor is benign or suspicious in a high surgical risk patient (although glutamine-based PET approaches have the potential to do this in the future; see previously), percutaneous renal biopsy can be safely performed with minimal risk of seeding (0.01% of cases), with high sensitivity (80%–100%) and specificity (83%–100%)^{135,137,138} and high negative predictive value (82%) and positive predictive value (97.5%) in one series of 2474 percutaneous biopsies.¹³⁷ The rare case of seeding have been reported in transitional cell cancers, which may carry a slightly higher risk¹³⁵; however, no case of seeding has been reported since 2001, particularly with the use of coaxial techniques, which avoid potential tumor exposure to the neighboring abdominal organs.^{137,138}

TUMOR STAGING AND GRADING

Tumor-node-metastasis (TNM) staging for RCC was first established in 1997 by the Union Internationale Contre le Cancer (UICC) and American Joint Committee on Cancer (AJCC) and was most recently updated in 2017 (Table 41.3).¹³⁹ T staging (T1–T4) is classified by the extent and size of the

tumor, which differentiate cancer-specific survival. T1 tumors are limited to the kidney and are 7 cm or smaller; T2 tumors are larger, more than 7 cm, but are also in the kidney; T3 tumors extend beyond the kidney but are within Gerota fascia and may involve neighboring veins (renal vein or inferior vena cava); and T4 tumors invade Gerota fascia or extend to the ipsilateral adrenal gland. T1 and T2 are further subdivided according to renal mass size, and T3 is subdivided depending on venous involvement.¹³⁹ T staging imposes the greatest discrimination of 5-year cancer-specific survival, where T1a tumors have a survival rate as high as 98%, which declines to 10% with stage T4, as shown in Table 41.3.¹²⁰ Nodal invasion minimally alters outcomes, but distant metastases, including ipsilateral adrenal gland invasion, worsen the prognosis considerably.¹²⁰ Renal sinus fat involvement has also been found to be associated with lower survival.¹⁴⁰

Composite prognostic staging (I–IV) summarizes the TNM findings. Tumors with stage I or II have kidney-limited lesions only. The prognostic stage is elevated to stage III when there is any nodal involvement, regardless of kidney size or T staging, and is escalated to stage IV when Gerota fascia invasion, adrenal gland, or distant metastasis occurs (Fig. 41.3).¹³⁹

Grading determines prognosis based on nuclear features. Of the grading systems, the International Society of Urologic Pathology (ISUP)–defined grading system (ISUP grading system) is the currently recommend grading system; however, for some time, the Fuhrman grade has universally been used for prognostic assessment, despite some limitations. The nuclear size, irregularity, and nucleoli prominence differentiate RCC into four grades. Fuhrman grade 1 tumors have small nuclei and no nucleoli; grade 2 tumors have larger irregular nuclei and the presence of nucleoli; grade 3 tumors also

Table 41.3 Tumor-Node-Metastasis Staging^a

Stage	Category	Criteria ^b	Tumor Invasion
Tumor	T0	No primary tumor	Confined to the kidney
	T1	Tumor ≤7 cm	
	T1a	Tumor ≤4 cm	
	T1b	Tumor 4–7 cm	
	T2	Tumor >7 cm	
	T2a	Tumor >7–10 cm	Confined to the kidney
	T2b	Tumor >10 cm	
	T3	Major veins or perinephric tissues	
	T3a	No ipsilateral adrenal gland	
		Renal vein and branches <i>or</i>	
		Renal pelvis and calyces <i>or</i>	
Node	T3b	Perirenal ± renal sinus fat	Confined within Gerota fascia
		Vena cava below the diaphragm	
		Vena cava above the diaphragm <i>or</i>	
	T3c	vena cava wall	
		± Ipsilateral adrenal gland	
	T4	± Ipsilateral adrenal gland	Extends beyond Gerota fascia
	N0	No lymph nodes	
	N1	Any regional lymph node(s)	
	M0	No distant metastasis	
	M1	Distant metastasis	

^aBy the American Joint Committee on Cancer (AJCC) and Union for International Cancer Control (UICC), 2017 update.

^bTumor size and perinephric tissue involvement.

Modified from Rini BI, McKiernan JM, Chang SS, et al., eds. *Kidney*. 8th ed. New York: Springer; 2017.

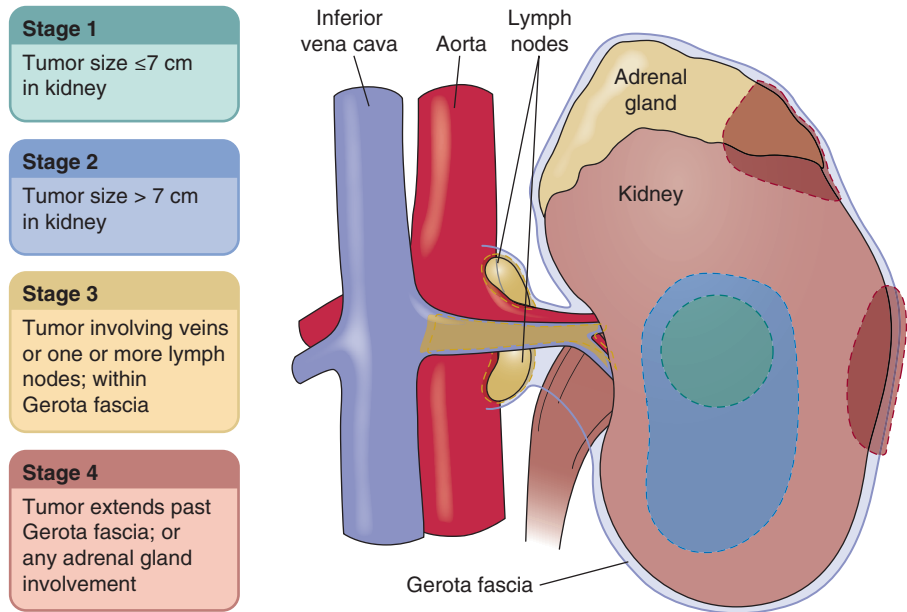


Fig. 41.3 The tumor node metastasis (TNM) staging classification. Shown is the general staging by the American Joint Committee on Cancer (AJCC) and Union for International Cancer Control (UICC), updated in 2017. (From Amin MB, Edge S, Greene F, et al., eds. *AJCC Cancer Staging Manual*. 8th ed. Springer International Publishing: American Joint Commission on Cancer; 2017 and Brierley JD, Gospodarowicz MK, Wittekind C, Eds. *TNM Classification of Malignant Tumours*, 8th ed. Wiley-Blackwell: 2017.)

Table 41.4. Five-Year Cancer-Specific Survival and Prognosis of Renal Cell Carcinoma^a

Prognosis	TNM Stage	Grade	Histologic Subtype	Sarcomatoid Differentiation	Rhabdoid Differentiation	Microscopic Coagulative Tumor Necrosis
Better	<i>Low stage</i> T1a, 98% T1b, 89% T2a, 75% T2b, 69%	<i>Low grade</i> 1: 51%–93% 2: 32%–86%	Papillary, 56%–92%; chromophobe, 78%–99%	Absence 79%	Absence	Absence
Worse	<i>High stage</i> T3a, 53% T3b, 31% T3c, 36% T4, 10%	<i>High grade</i> 3: 25%–50% 4: 10%–28%	Clear cell, 50%–86%	Presence, 15%–22%	Presence	Presence

^aDetermined by staging, grading, and other histologic features.
TNM, Tumor-node-metastasis.

have large irregular nuclei, but with more visible nucleoli; and grade 4 tumors have extremely abnormal nuclei and highly prominent nucleoli apparent at low power.¹⁴¹ The 5-year cancer-specific survival is highest for grade 1 and worst in grade 4, as illustrated in Table 41.4.¹²⁰ In some studies, Fuhrman grades 2 and 3 are thought to differentiate outcomes poorly.¹²⁰ Although this grading system has been reliably used for ccRCC, it has not been adequately validated for papillary or chromophobe RCC subtypes.^{140,142} In 2012, the ISUP grading system further classified nuclear findings at specific magnifications and expanded criteria to include rhabdoid and sarcomatoid features (Fig. 41.4).¹⁴³

Independent of TNM staging, higher grade and larger tumor size predict poorer survival. Histologic subtype is also a factor in survival. The most common and sporadic form of RCC, clear cell cancer (60%–90% prevalence vs. 6%–14% for papillary and 6%–14% for chromophobe), has inferior

outcomes than those observed with papillary or chromophobe RCC. The 5-year survival for papillary and chromophobe RCC range is higher than that for ccRCC.¹⁴² Other features, such as the presence of sarcomatoid or rhabdoid (eccentric large nuclei and eosinophilic cytoplasm) differentiation, now included in the ISUP grading system, and microscopic coagulative tumor necrosis add to the lower prognosis. Sarcomatoid differentiation, seen in only 4% of all RCC, reflects extreme dedifferentiation and is associated with a dismal prognosis. Each 10% increase of sarcomatoid changes is associated with a 6% increased risk of death from RCC, with less than a 1-year mean survival. Rhabdoid differentiation is even rarer, with similarly poor survival.^{120,140}

Various integrated staging systems based on variables, often including TNM staging, tumor size, clinical symptoms, histologic subtype, grading, and other prognostic pathologic findings, can be used to predict outcomes preoperatively

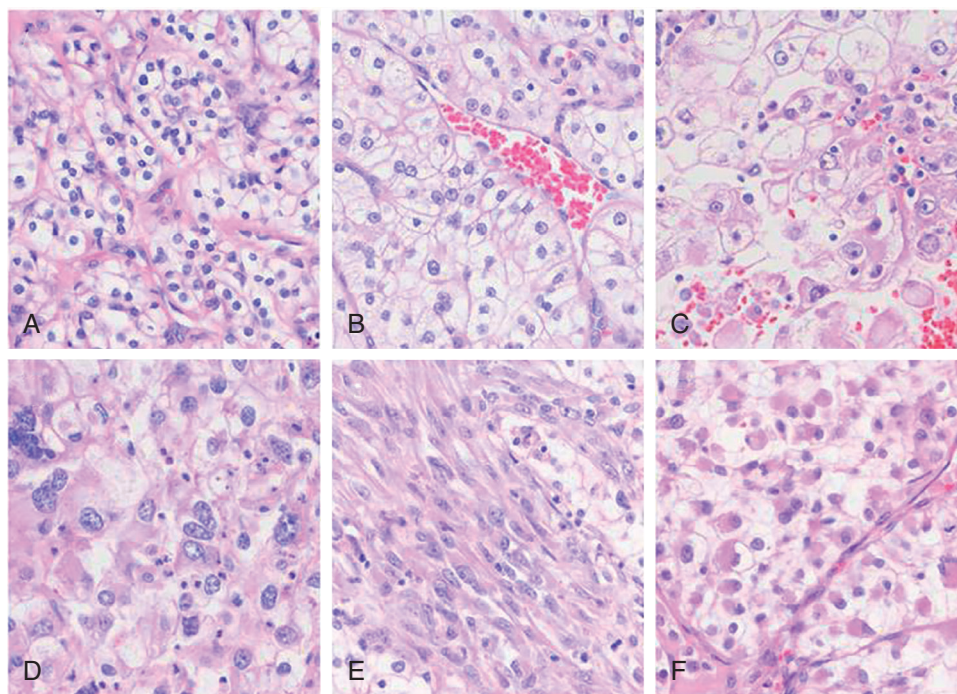


Fig. 41.4 The International Society of Urologic Pathology (ISUP) grading system for clear cell and papillary renal cell carcinoma (RCC). (A) Grade 1: Nucleoli are inconspicuous or absent. (B) Grade 2: Nucleoli are clearly visible at high-power magnification but are not prominent. (C) Grade 3: Nucleoli are prominent and are easily visualized at low-power magnification. (D) Grade 4: Presence of tumor giant cells and/or marked nuclear pleomorphism. (E) Grade 4: Sarcomatoid carcinoma. (F) Grade 4: Carcinoma showing rhabdoid differentiation.

and postoperatively. The accuracy of these predictions is high for RCC recurrence (66%–80%), distant metastases (78%–85%), and RCC cancer-specific survival (64%–89%).¹⁴⁴ Such algorithms are particularly helpful for patients with advanced disease (metastatic RCC) who are in the highest mortality range and therefore are poorly differentiated by TNM staging alone. Even the therapeutic effectiveness of antineoplastic treatments (cytokine or targeted therapy) has been examined using these staging systems to select optimal treatment regimens.^{120,144}

TREATMENT

The role of the nephrologist in the management of renal masses is evolving, because more than 50% of RCCs are being diagnosed incidentally and earlier with advanced diagnostic imaging modalities.^{145,146} Although a proportion ($\approx 20\%$) of these masses tends to be nonmalignant lesions, proper diagnosis often still requires surgical excision.^{147,148} Fortunately, localized disease has a 5-year survival as high as 100%, particularly for those with small renal masses (T1a tumors ≤ 4 cm). Survival in this group is not driven entirely by cancer-related factors, but rather by complications related to their RCC treatment or comorbid diseases, which may result in CKD or cardiovascular disease.^{149–151} The risk of death among those with advanced disease, however, remains high with different prognostic factors. Therefore, therapy diverges into two arms depending grossly on whether the tumor is localized or advanced. Localized disease is managed surgically, with greater emphasis on preserving kidney function, whereas advanced disease requires systemic treatment with the goal to maximize cancer-specific survival.⁴

LOCALIZED RENAL MASSES

PREOPERATIVE CONSIDERATIONS: CHRONIC KIDNEY DISEASE BURDEN

Shared risk factors for CKD and RCC—older age, tobacco exposure, DM, HTN, and obesity—in addition to a high prevalence of CKD in the small renal mass (≤ 4 cm) population, predisposes these individuals to an increased risk of new-onset CKD, CKD progression, or ESRD after surgical resection.^{8,70} CKD is present in 10% to 30% of patients with RCC,^{152–154} but the prevalence is even higher (10%–52%) when considering only those with small renal masses.^{155–158} The higher burden of CKD risk factors in the small renal mass population (e.g., nearly 25% with DM and up to 50% with HTN) likely contributes to rising CKD incidence.^{154,155,157,159,160} RCC patients were more likely than case-matched, non-RCC patients to have DM (20% vs. 8%) and HTN (31% vs. 14%) in a cohort of 26,460 patients in Taiwan.¹⁶¹ Furthermore, given the increased age among those with RCC, more are likely to be saddled with preexisting CKD.^{153,156}

POSTOPERATIVE CHRONIC KIDNEY DISEASE RISK

Treatment of localized renal masses generally results in reduction of a functional nephron mass, which can lead to a decline in the eGFR. In addition to the higher burden of preexisting CKD and risk factors for CKD, as noted earlier,^{162–164} the level of eGFR and poor nutrition (lower albumin), as well as surgical factors, including tumor size and acute kidney injury, has been associated with a higher incidence of CKD posttreatment^{156,157,161,165–167} (Fig. 41.5). The prevalence of postnephrectomy CKD has doubled from 10% to 24% to 16% to 52% in multiple cohorts,^{155–157} and

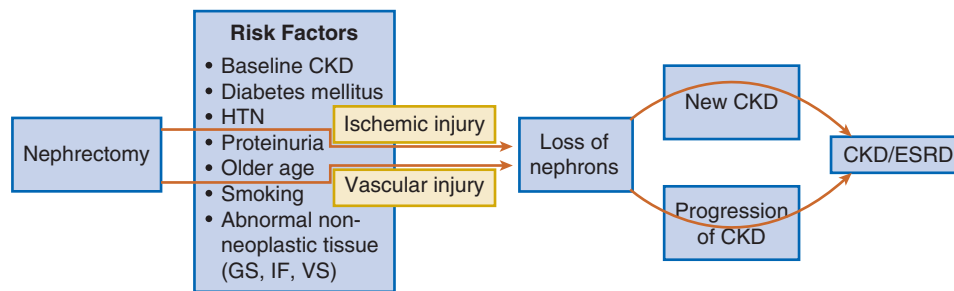


Fig. 41.5 Risk factors for chronic kidney disease (CKD). Nephrectomy performed for renal cell carcinoma is associated with nephron loss due to tissue removal, as well as ischemic and vascular injury. This loss of nephrons is associated with either new-onset CKD or progression of CKD in patients who have various risk factors, as noted. *ESRD*, End-stage renal disease; *GS*, glomerulosclerosis; *IF*, interstitial fibrosis; *HTN*, hypertension; *VS*, vascular sclerosis.

the eGFR decreased by 30% (mean, 13 mL/min/m² reduction) when followed longitudinally after nephrectomy.¹⁶⁷ Having DM as a comorbidity raised this risk significantly among those with small renal masses after nephrectomy, resulting in new-onset CKD, 60% in diabetics compared with 43% among all-comers with RCC. In this cohort, only 46% of diabetics were likely to remain CKD-free after 2 years compared with 76% of nondiabetics ($P = .006$).¹⁵⁷ For ESRD, in a case-control study of 26,460 Taiwanese patients with 10-year follow-up, the incidence rate was 4.05% for the RCC group ($n = 2940$) compared with 0.68% in the control group ($n = 23,520$), with a 5.6 times more likely risk of ESRD (HR, 5.6; CI, 4.3–7.2).¹⁶¹ Ultimately, RCC has been determined to be a prominent cause of ESRD in 0.5% of the population (360,000 patients), according to the US Renal Data System (USRDS).¹⁶⁴

RENAL PATHOLOGY

Although the primary purpose of pathologic evaluation is the management of RCC, proper identification of renal pathology beyond the neoplastic tissue is vital, because patients with small renal masses are outliving their cancer and, instead, dying from other diseases such as CKD and its complications including cardiovascular disease. The College of American Pathologists has recommended concomitant examination of nonneoplastic pathology, but compliance has been suboptimal.^{168–170} As we have recently proposed,⁸ a multidisciplinary team of the urologist, nephrologist, and pathologist should aim to achieve accurate diagnosis of the noncancerous renal disease (e.g., glomerular, tubulointerstitial, or vascular) to facilitate treatment of the underlying disease and promote GFR preservation.

SURGICAL TREATMENT

PREOPERATIVE CLINICAL EVALUATION

Before surgery or an alternative nephron-sparing procedure, risk assessment for CKD should be performed. Identification of preexisting CKD (serum creatinine level and eGFR calculation) and urinary proteinuria assessment (urine albumin-to-creatinine or urine protein-to-creatinine ratio) further allows for CKD staging using the Kidney Disease Improving Global Outcomes (KDIGO) classification criteria.^{171,172} Preservation of kidney function may be achieved by addressing CKD risk factors prior to RCC intervention.

Optimization of comorbid diseases encompassing glycemic control among diabetics, blood pressure adjustment for those with HTN, prevention of acute kidney injury through medication review, and avoidance of hypotension and nephrotoxicity have been recommended.^{4,173,174} Further characterization of differential kidney function with renal nuclear scintigraphy prior to nephrectomy may help prognosticate the risk of CKD after treatment; however, these studies are limited because they tend to underestimate the postoperative function of the remaining kidney, which is often augmented by hyperfiltration and hypertrophy.¹⁷⁵ Furthermore, surgical factors, including renal mass size and duration of ischemia, correlated better with postoperative kidney function than renal functional scans.¹⁷⁶

RENAL MASS EXCISION AND NEPHRON-SPARING THERAPIES

For decades since the first nephrectomy in 1861, the primary treatment of kidney cancers had been radical nephrectomy, which involved total removal of the kidney and surrounding tissues, such as the adrenal glands or lymph nodes. Although partial nephrectomies had been performed not too long after radical nephrectomy (1867), they were generally indicated for nonmalignant lesions and did not gain acceptance until well into the 21st century^{177,178}; this procedure is still not routinely performed in all centers.¹⁷⁹

Nephron-sparing procedures for small renal masses have proven to be safe with comparable, if not better, outcomes than those for radical nephrectomy in numerous studies, especially with regard to CKD. A meta-analysis of 40,000 subjects in 36 studies composed of primarily retrospective cohort studies and one randomized-controlled trial (RCT) has summarized these findings. The composite all-cause mortality (19%) and cancer-specific mortality (29%) were lower in the partial nephrectomy group than in the radical nephrectomy group.¹⁸⁰ These findings, driven by the large number of cohort studies, differed from the one RCT, the European Organisation for Research and Treatment of Cancer (EORTC) study, which assessed 541 patients with renal masses 5 cm or smaller. The EORTC trial did not support higher overall survival with partial nephrectomy; the 10-year all-cause mortality was 81% for radical nephrectomy versus 76% for partial nephrectomy (HR; 1.5; 95% CI, 1.03–2.16). However, the difference in survival between radical nephrectomy and partial nephrectomy was no longer present when the analysis was restricted to the subgroup with a ccRCC diagnosis.¹⁵⁰

Kidney function preservation in those with small renal masses is increasingly prioritized as oncologic survival approaches 100%. The CKD risk was 61% less with partial nephrectomy as compared with radical nephrectomy in the meta-analysis noted earlier.¹⁸⁰ Reduction of CKD risk may partially explain the lower all-cause mortality, because it has been well established that CKD is associated with cardiovascular death.¹⁸¹ Likewise, eGFR decline was also lower for partial nephrectomy in the EORTC study. Compared with 86% of patients undergoing radical nephrectomy, fewer (65%) with partial nephrectomy had an eGFR decline below 60 mL/min/1.73 m². However, there was no distinction of CKD benefit between these groups when examining the risk of steeper eGFR declines (<30 or 15 mL/min/1.73 m²) according to surgical modality. Despite a higher mean eGFR (67 vs. 53 mL/min/1.73 m²) among those with partial nephrectomy, this did not appear to confer a survival benefit in this trial.¹⁸² The eGFR reduction related to surgical nephron mass loss may not raise mortality like long-standing CKD.¹⁸³ Furthermore, although ESRD did not occur more frequently with radical nephrectomy than with partial nephrectomy (in a large Canadian study spanning 1995 to 2010), the risk of ESRD for those undergoing partial nephrectomy during a contemporary period (2003–2010) was lower (HR, 0.44; 95% CI 0.25–0.95) compared with radical nephrectomy. This is presumably related to the advancement of surgical practice for the former group creating disparate outcomes over time.¹⁸⁴

Ablative therapies and active surveillance for the treatment of small renal masses should be considered for those who may be excluded from nephrectomy (radical or partial nephrectomy) due to the high risk of surgical complications. Radiofrequency ablation and cryoablation are well-established nephron-sparing procedures usually reserved for older adults and those with multiple comorbid conditions. These less invasive ablative therapies have the advantage of similar oncologic survival, lower procedural complication rates, fewer hospitalization days, and greater GFR preservation, but with more local and metastatic tumor recurrence compared with partial nephrectomy.^{185,186} Active surveillance entails intensive and consistent monitoring of the tumor, with surgical intervention, as indicated. Compared with more aggressive measures, which start with surgical or ablative treatment, active surveillance performs well in select populations (as observed with older adults in the U.S. Surveillance Epidemiology and End Results [SEER] database)¹⁸⁷ and those with lower life expectancy or competing risks of death.^{187,188} Surgical outcomes are similar but oncologic survival is slightly lower among those undergoing active surveillance.¹⁸ The American Urological Association and American Society of Clinical Oncology guidelines have recommended active surveillance for those with masses 2 cm or smaller or with high operative risk, with monitoring every 3 to 6 months.^{174,190} Moreover, a few prospective observational studies of those with metastatic disease (with favorable prognostic features) have found that a period of active surveillance (median, 1 year) prior to systemic therapy may possibly minimize toxicity related to treatment.¹⁹¹

POSTTREATMENT SURVEILLANCE OF LOCALIZED TUMORS

The American Urological Association has recommended regular screening for tumor recurrence by imaging (abdominal

CT, MRI, or ultrasound and chest radiography) initially after treatment. Imaging is repeated between 3 and 12 months postprocedure and thereafter yearly, up to 3 (partial nephrectomy) and 5 (ablative therapy) years, depending on the treatment modality. Among those who received ablative therapy, renal biopsy is indicated during surveillance with the discovery of concerning lesions. Anyone with preexisting or a new diagnosis of CKD (by eGFR or proteinuria criteria) should be monitored regularly for progression of kidney disease.^{150,192}

ADVANCED RENAL CELL CARCINOMA AND SYSTEMIC THERAPY

Although incidental renal mass detection has led to earlier RCC diagnosis, with favorable outcomes, nearly one-third of RCC patients continue to be diagnosed with metastatic disease or recurrent disease that is not amenable to surgical therapy. Instead, systemic therapy is the mainstay of treatment for advanced RCC, although surgical debulking may be indicated for reducing tumor burden in some cases. Advanced RCC median survival remains suboptimal; however, with the development of newer treatment options, prognoses appear to be improving.^{193,194}

INTRODUCTION TO NEW THERAPEUTIC APPROACHES

Disruption of VHL through mutation, deletion, or methylation is widely recognized as the most fundamental and critical molecular alteration in ccRCC.¹¹ This disruption leads to the deregulation of downstream pathways related to oxygen sensing, causing the accumulation of HIF- α and subsequent upregulation of hypoxia response genes,¹⁹⁵ and resulting in a clinical phenotype characterized by florid vasculature, principally modulated by VEGF signaling pathways. In the past decade, remarkable advances have been made in ccRCC drug development, primarily focusing on targeting either VEGF receptors (VEGFRs) using kinase inhibitors or the VEGF ligand using monoclonal antibodies. Such advances have led to FDA approval of several inhibitors of VEGFR (sunitinib, pazopanib, sorafenib, axitinib, cabozantinib, and lenvatinib) and VEGF (bevacizumab). These angiogenesis inhibitors were only modestly effective, however, and were often characterized by off-target effects and chronic irritative toxicities, such as fatigue and rash.¹⁹⁶ Furthermore, the metabolic basis of ccRCC is only partly addressed by targeting the terminal phenotype driven by VEGF. Subsequent encouraging progress in drug development efforts has attempted to exploit ccRCC metabolic reprogramming by targeting critical proteins or enzymes involved in dysregulated metabolic pathways.^{3,23}

IMMUNOTHERAPY

Interferon- α (IFN- α) and IL-2 were the initial immune therapies to be approved by the FDA. Cytokine therapy stimulates the immune system nonspecifically with the upregulation of T-cell and natural killer cell activity to recognize and destroy tumor cells,¹⁹⁷ but can also have the undesired effect of suppressing the immune system

and defeating natural immunosurveillance.^{24,40} Response rates have ranged roughly from 10% to 25% and median overall survival from 11 to 13 months for individual agents, without much better overall survival for combination therapy. Immunotherapy, when tolerated, is reasonably successful but is severely limited by the frequent withdrawal of treatment due to the flulike adverse effects experienced in most patients.¹⁹³

TARGETED MOLECULAR THERAPY

Tumor growth and angiogenesis are mediated through a number of intracellular pathways involving VEGF, HIF, and mammalian target of rapamycin (mTOR; Fig. 41.6). Treatments targeting these pathways generally inhibit the enhanced tumor cellular metabolism, cell growth, and vascular network expansion and have revolutionized the approach to cancer therapy in all solid organs, including the kidney.

VASCULAR ENDOTHELIAL GROWTH FACTOR

The first VEGF inhibitor (bevacizumab) targets the VEGF ligand, preventing angiogenesis by the reduction of protein transcription, which promotes vascular growth and increasing oxygen delivery. VEGF inhibitors also impede tumor cell growth directly because it is present not only in endothelial cells but also in many other cell types, including tumor cells. Presently, VEGF activity is also blocked at the receptor using tyrosine kinase inhibitors (TKIs), including axitinib,

pazopanib, sorafenib, sunitinib, and tivozanib.^{198,199} Although highly effective, resistance can occur from the stimulation of alternative angiogenic pathways bypassing the VEGF blockage.²⁰⁰

Common adverse effects of VEGF inhibitors, which are important for the nephrologist to know and treat, are HTN (11%–43%)²⁰¹ and proteinuria (41%–63%)²⁰² related to the disruption of the role of VEGF in maintaining the integrity of the endothelium and slit diaphragm of the glomerular membrane. On pathology, the primary presentation is thrombotic microangiopathy, but various glomerular diseases have been reported as well.^{203,204} The pathogenesis mirrors disease states of VEGF antagonism, as seen with preeclampsia.²⁰⁵ The risk of HTN and proteinuria appears to increase with larger VEGF inhibitor dose.^{201,202} The drug toxicity is generally treated supportively with antihypertensive agents and protein-lowering medications, such as angiotensin-converting enzyme inhibitors (ACEIs) and angiotensin receptor blockers (ARBs). Severe HTN or nephrotic syndrome resistant to medical therapy can be addressed with short-term cessation or lowering of the dose, without interruption of therapy, except for severe acute kidney injury or thrombotic microangiopathy.²⁰⁶

PI3K/MTOR

Pathways involving mTOR and its upstream serine–threonine kinase, phosphatidylinositol-3-kinase (PI3K), also promote

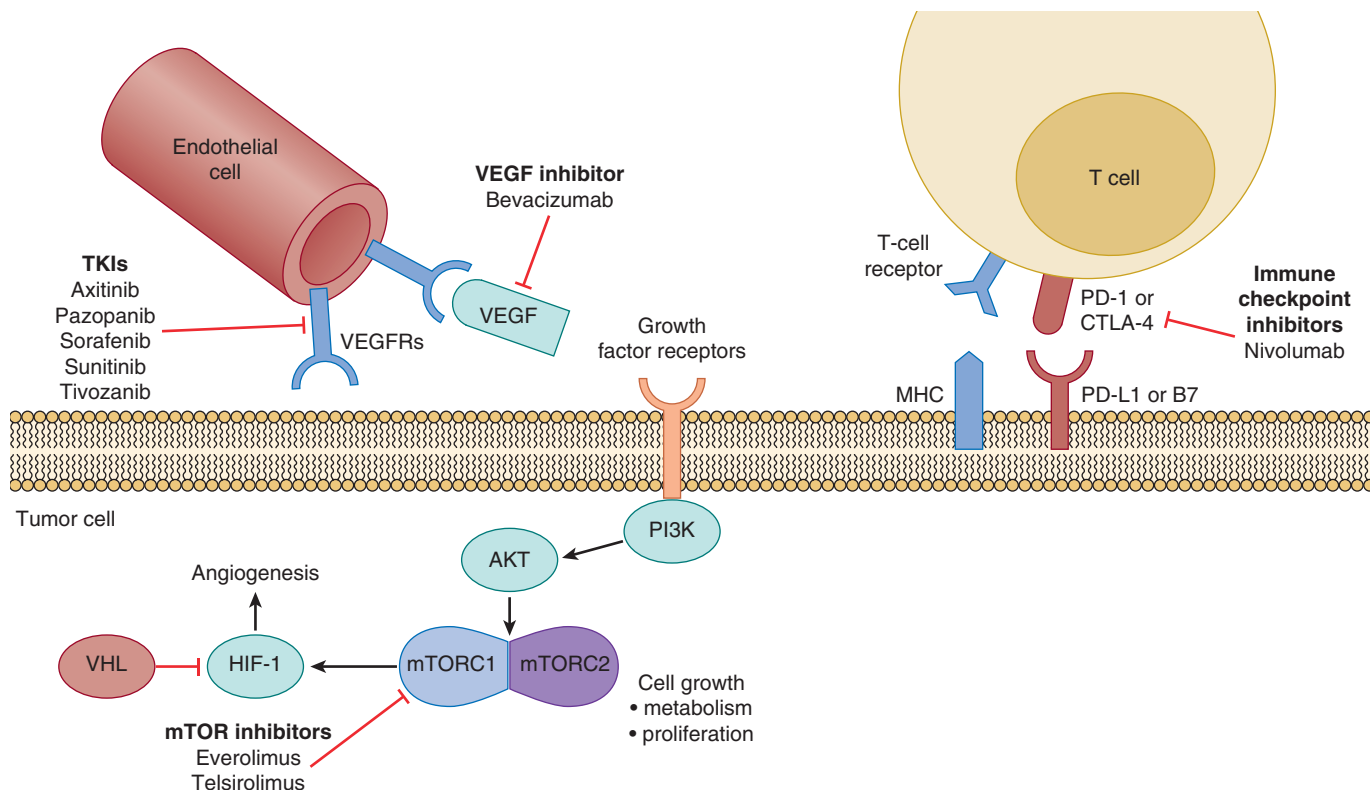


Fig. 41.6. Tumor growth, angiogenesis, immune pathways, and associated targeted immune therapy. Tumor cells promote growth and angiogenesis through the PI3K/mTOR and VEGF pathways and evade the T-cells and immune checkpoints. Various targeted immune therapies are listed; the site of action is indicated by the red line. B7, CD-28 ligand; CTLA-4, cytotoxic-T-lymphocyte-associated antigen-4; HIF-1, hypoxia-inducible factor-1; MHC, major histocompatibility complex; MTOR, mammalian target of rapamycin; mTORC, mammalian target of rapamycin complex; PD-1, programmed cell death protein-1; PD-L1, programmed cell death protein-1 ligand 1; PI3K, phosphatidylinositol-3-kinase; TKI, tyrosine kinase inhibitor; VEGF, vascular endothelial growth factor; VEGFR, VEGF receptor; VHL, von Hippel-Lindau.

tumor growth. mTOR is crucial for ribosomal activity in cell metabolism, growth, and proliferation through two separate complexes (mTORC1 and mTORC2), and the PI3K pathway controls growth factor expression and tumor cell proliferation. The reported response rate is 7% to 26%, with a higher median overall survival of 15 months compared with traditional immunotherapy, and these agents have a far more tolerable toxicity profile. The first-generation mTOR inhibitors (everolimus, temsirolimus) suppress mTORC1 activity alone; therefore, mitogenic stimulation can occur alternately through mTORC2 and PI3K. The next generation of mTOR inhibitors also block the mTORC2 and PI3K pathway, which may allow for more effective tumor suppression, and is currently undergoing investigation.^{4,193,200}

NOVEL TARGETED THERAPIES

Immune checkpoint receptors sustain normal autoregulation of our immune system by downregulating T-cell and associated immune cell (macrophage, natural killer cell) activity. Tumors cells “hijack” this function to evade circulating immune cells. Immune checkpoint inhibitors block the aberrant tumor to T-cell interaction, enabling T cells to recognize the deviant cells and stimulate the natural immune mechanisms to eradicate them. Cytotoxic T lymphocyte-associated antigen-4 (CTLA-4) and programmed cell death protein-1 (PD-1) are two receptor proteins regulating such adaptive immunity used in cancer treatment. Ipilimumab (CTLA-4-directed antibody) and nivolumab (PD-1-directed antibody) have been used successfully in the treatment of RCC.²⁰⁷ The response rate for nivolumab (25%) has been respectable, although with a rare complete response; median overall survival has been reported as high as 25 months.²⁰⁸ Ipilimumab appeared less effective, with a response rate of only 12.5% in one high-dose phase II trial.²⁰⁹ Assessment of combination therapy is ongoing, with optimistic results so far.²⁰⁷ Toxicity related to checkpoint inhibitors tends to present with cutaneous, gastrointestinal, pulmonary, and hepatic involvement. Less commonly, renal complications can occur, with acute kidney injury due to acute tubulointerstitial nephritis (lymphocytic infiltration), which typically resolves with corticosteroid administration or cessation of treatment.²¹⁰ Further investigation will help determine optimal treatment regimens for those with advanced RCC.

In most patients with metastatic renal cell cancer of the clear cell type, systemic VEGF-targeted therapy (e.g., sunitinib, pazopanib, bevacizumab/interferon) is considered a reasonable frontline standard of care. Although the mTOR inhibitor temsirolimus is also used to treat patients with a poor functional status, many of these patients are also reasonable candidates for VEGFR tyrosine kinase inhibitor (TKI) therapy. In highly selected patients with a robust functional status, high-dose IL-2 remains a treatment consideration due to its ability to induce durable responses in a small subset of patients. In the second-line setting, systemic therapy with immune checkpoint inhibitors (nivolumab) or TKIs (e.g., cabozantinib, axitinib, lenvatinib) are in the therapeutic armamentarium. The mTOR inhibitor everolimus has now been relegated to later lines of therapy.^{194,211}

The use of -omics technologies to discover and validate reprogrammed metabolic pathways is starting to have an impact on RCC therapy.^{18,19,40} This idea, which is especially germane to the metabolic basis of RCC, capitalizes on the

availability of novel or previously discovered compounds that affect such pathways and thus are expected to affect cancer preferentially over normal tissues.³ We foresee a bright future of newly discovered therapeutic targets using this technology, which could be rapidly and easily translated to the clinical setting.



Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

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BOARD REVIEW QUESTIONS

1. The risk of renal cell carcinoma (RCC) is greatest in which of the following patients?
- A 36-year-old man with regular alcohol use and polycystic kidney disease
 - A 46-year-old woman with coronary artery disease and congestive heart failure
 - A 70-year-old man with end-stage renal disease on dialysis for 10 years
 - A 65-year-old female renal transplant recipient

Answer: c

Rationale: In general, older men in their sixth to seventh decade of life with diabetes mellitus, hypertension, tobacco exposure, obesity, and end-stage renal disease, with prolonged dialysis, are at increased risk for RCC. Cystic diseases such as acquired cystic kidney disease is associated with RCC, but polycystic disease is not.

2. Which of the following statements regarding chronic kidney disease (CKD) and renal cell carcinoma is true?
- Radical nephrectomy is associated with a lower risk of CKD.
 - Routine preoperative screening with urinalysis for proteinuria is recommended.
 - Kidney function assessment with creatinine is not recommended for those with small renal masses.
 - Tumor size does not determine the risk of CKD.

Answer: b

Rationale: Shared risk factors for CKD and RCC (older age, tobacco exposure, diabetes mellitus, hypertension, and obesity), in addition to a high prevalence of CKD in the small renal mass population (≤ 4 cm), predisposes them to an increased risk of new-onset CKD, CKD progression, or end-stage renal disease (ESRD) after surgical resection. In addition, the level of estimated glomerular filtration rate (eGFR), poor nutrition (lower albumin), and surgical factors, including tumor size and acute kidney injury, have been associated with a higher incidence of posttreatment CKD. Identification of preexisting CKD (serum creatinine and eGFR calculation) and proteinuria assessment (urine albumin-to-creatinine or urine protein-to-creatinine ratios) further allows for CKD staging using the Kidney Disease Improving Global Outcomes (KDIGO) classification criteria, as recommended by the American Urological Association guidelines. CKD risk is less with partial nephrectomy as compared with radical nephrectomy.

3. Renal cell carcinoma is called “the internist’s tumor” because
- It occurs in the internal compartment of the body.
 - Primary care internists are the most likely to treat this disease.
 - RCC has multiple and varied presentations that require the astute mind of an internist to put together and make the diagnosis.

Answer: c

Rationale: Clear cell RCC, the most common type, is a metabolic disease associated with many abnormalities in metabolic pathways. Thus, because each pathway abnormality results in a discrete sign or symptom that often results from cytokine or aberrant hormone production, ccRCC often presents in a pleiotropic fashion and requires the full intellectual force of a good internist for diagnosis.

4. RCC has been notoriously difficult to treat due to tumor resistance after several years. New approaches to treatment include which of the following?
- Immune checkpoint inhibitors
 - Interferon-based therapies
 - Kinase inhibition

Answer: a

Rationale: Interferon-based therapies have been successful for only a small fraction of cases, and kinase inhibitors such as sorafenib and sunitinib have been associated with tumor resistance. Immune checkpoint inhibitors are showing promise, especially with combination therapy.

5. Which of the following statements concerning the genetics of ccRCC is true?
- Most cases of ccRCC are linked to hereditary diseases.
 - Mutations of the von Hippel-Lindau (*VHL*) gene are strongly linked with sporadic ccRCC.
 - VHL disease, although linked to ccRCC, is extremely rare.
 - Mutations of the *VHL* gene are the only genetic abnormalities linked to ccRCC.

Answer: b

Rationale: Although VHL disease is relatively common and associated with ccRCC, most cases of ccRCC are not linked to hereditary diseases. However, mutations of the *VHL* gene are found in more than 80% of cases of sporadic ccRCC. Other genetic abnormalities associated with ccRCC include mutations in epigenetic regulators and chromatin remodeling genes and disruption of the PI3K-AKT-mTOR signaling pathways.